SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: MLA0304
COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO5: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 35%

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QUESTIONS: 10 **Total Marks: 50**

Annexure A

Scenario-Based Assessment **DURATION: 1.5 Hrs**

1. A mobile app company wants to improve user engagement by showing personalized notifications at the right time. The company plans to use Reinforcement Learning to learn user behavior patterns. Report how the RL framework can be applied by defining the possible states, actions, and reward functions for this scenario. **(5 Marks)**

2. A warehouse robot uses Reinforcement Learning to find the shortest path for delivering packages while avoiding obstacles. Sometimes the robot chooses longer paths due to insufficient exploration during training. Solve how the exploration vs exploitation trade-off affects the robot's navigation performance and suggest some improvements. **(5 Marks)**

3. A hospital is considering using Reinforcement Learning to optimize patient appointment scheduling in order to reduce waiting time and improve resource utilization. Demonstrate the suitability of RL for this scenario and discuss possible challenges such as delayed rewards and ethical considerations. **(5 Marks)**

4. A smart home system aims to reduce electricity consumption by automatically controlling lights, fans, and air conditioners based on user presence and weather conditions. Sketch a Reinforcement Learning model for this scenario by defining state space, action space, reward structure, and suitable learning algorithm.

(5 Marks)

5. An online gaming platform uses Reinforcement Learning to match players of similar skill levels to maintain engagement. Report how reward design can influence matchmaking quality with reward definition that might lead to biased or inefficient matching results. **(5 Marks)**

6. A financial trading system applies Reinforcement Learning to decide buy/sell actions based on market signals. Analyze how states, actions, and rewards can be defined in this context and evaluate the risks of delayed rewards in volatile markets.

(5 Marks)

7. A drone delivery company uses RL to optimize delivery routes while avoiding restricted airspace. Construct a suitable RL framework by defining states, actions, and rewards, and justify how penalties can ensure safe navigation. **(5 Marks)**

8. A university applies RL to personalize online learning paths for students. Examine how reward design can influence student engagement and defend the ethical considerations of adapting learning content dynamically. **(5 Marks)**

9. A traffic management authority deploys RL to control traffic lights at busy intersections. Illustrate how constraints such as emergency vehicle priority and pedestrian safety can be incorporated into the RL model and interpret the impact on congestion reduction. **(5 Marks)**

10. A manufacturing plant uses RL to schedule machines under resource limitations. Differentiate between traditional optimization methods and constrained RL approaches, and assess which method provides better adaptability in dynamic production environments. **(5 Marks)**

Code of Conduct:

*I_(*K Varshan / 192325089*) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

K Varshan
Signature of the student:

ANSWERS:

1. Personalized Notifications Using Reinforcement Learning

Understanding of Scenario

The mobile app wants to increase user engagement by sending personalized notifications at suitable times. Reinforcement Learning can learn from the user's responses and gradually identify the best notification strategy.

State

The state can include:

- User activity level
- Time of day
- Previous notification response
- User's recent app usage
- Type of content preferred by the user

Actions

The RL agent can:

- Send a notification
- Delay the notification
- Select different notification content
- Choose different notification times

Reward

A positive reward can be given when the user opens or interacts with the notification. A negative reward can be given when the user ignores the notification or disables notifications.

Decision

RL is suitable because user behavior changes over time. The agent can continuously learn which notification strategy produces better engagement while avoiding excessive notifications.

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2. Exploration vs Exploitation in Warehouse Robot Navigation

Understanding of Scenario

The warehouse robot must find short paths while avoiding obstacles. The problem occurs when the robot does not explore enough and repeatedly uses paths that it already knows.

Key Issue

Exploration means trying new paths to discover better routes. Exploitation means selecting the best-known path. Too much exploitation can make the robot miss shorter paths, while too much exploration can cause unnecessary movement and delay.

Application

An epsilon-greedy strategy can be used. With probability ε, the robot selects a random action for exploration. Otherwise, it selects the action with the highest learned value.

Improvements

- Start with a higher exploration rate.
- Gradually reduce ε during training.
- Give higher rewards for shorter paths.
- Give strong negative rewards for collisions.
- Use experience replay or improved exploration methods.

Decision

A gradually decreasing exploration rate provides a good balance. Early exploration helps discover efficient routes, while later exploitation allows the robot to use the best learned path.

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3. Reinforcement Learning for Hospital Appointment Scheduling

Understanding of Scenario

The hospital wants to schedule patient appointments to reduce waiting time and improve the utilization of doctors, rooms, and other resources.

State

The state can include:

- Number of waiting patients
- Doctor availability
- Room availability
- Appointment queue length
- Patient priority
- Current time

Actions

The RL agent can assign, delay, reschedule, or prioritize appointments.

Reward

A positive reward can be given for reducing waiting time and efficiently using resources. A negative reward can be given for excessive waiting, resource conflicts, or missed appointments.

Key Challenges

Delayed rewards are a major challenge because the effect of a scheduling decision may only become visible after several appointments. Ethical issues must also be considered, particularly fairness and equal treatment of patients.

Decision

RL can be suitable for dynamic scheduling, but it should operate under hospital policies and fairness constraints. Human supervision should remain important for critical medical decisions.

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4. Smart Home Energy Management Using RL

Understanding of Scenario

The smart home wants to reduce electricity consumption by automatically controlling lights, fans, and air conditioners according to user presence and environmental conditions.

State Space

The state can contain:

- User presence
- Temperature
- Weather condition
- Current electricity consumption
- Time of day
- Appliance status

Action Space

The agent can:

- Turn appliances ON or OFF
- Increase or decrease AC temperature
- Adjust fan speed
- Adjust lighting level

Reward Structure

Positive rewards can be given for maintaining user comfort with low energy consumption. Negative rewards can be given for excessive electricity usage or uncomfortable temperature conditions.

Learning Algorithm

Q-Learning can be used for a small discrete environment. For a larger continuous environment, Deep Q-Network or another suitable RL algorithm can be considered.

Decision

RL is appropriate because the system can continuously adapt appliance control to changing user behavior and weather conditions while minimizing energy consumption.

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5. Reward Design in Online Gaming Matchmaking

Understanding of Scenario

The gaming platform wants to match players with similar skill levels so that games remain competitive and enjoyable.

Key Issue

The reward function strongly influences the quality of matchmaking. If the reward is poorly designed, the RL agent may learn a strategy that increases one metric while reducing fairness or player satisfaction.

Suitable Reward

The reward can consider:

- Skill similarity
- Balanced win probability
- Player engagement
- Game completion
- Player satisfaction

A positive reward can be given for balanced matches, while penalties can be applied to large skill differences and repeated mismatches.

Biased Reward Example

If the system gives rewards only for increasing playing time, it may repeatedly match players in ways that maximize engagement but create unfair or frustrating games.

Decision

The reward should combine engagement, fairness, and match quality. This prevents the agent from optimizing only one factor and producing biased or inefficient matchmaking.

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6. Reinforcement Learning for Financial Trading

Understanding of Scenario

The trading system uses RL to decide whether to buy, sell, or hold based on changing market information.

State

The state can include:

- Stock price
- Price movement
- Trading volume
- Technical indicators
- Market trends
- Current portfolio position

Actions

The agent can:

- Buy
- Sell
- Hold

Reward

The reward can be based on portfolio return or profit after considering transaction costs and risk.

Delayed Rewards

Trading decisions may not produce immediate rewards. A decision to buy today may produce a profit or loss several steps later. Market volatility can make this relationship even more difficult to learn.

Risk Analysis

Delayed rewards can cause incorrect credit assignment. Sudden market changes can also make previously learned strategies ineffective. Therefore, risk penalties and transaction costs should be included.

Decision

RL can be useful for adaptive trading strategies, but it should be evaluated carefully because financial markets are uncertain and highly volatile. Risk management should be included in the reward design.

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7. RL-Based Drone Delivery with Restricted Airspace

Understanding of Scenario

The drone delivery system must find efficient routes while avoiding restricted airspace and unsafe areas.

State

The state can include:

- Current drone position
- Destination
- Battery level
- Weather condition
- Nearby obstacles
- Restricted airspace information

Actions

The drone can:

- Move forward
- Move backward
- Move left
- Move right
- Move upward or downward

Reward

A positive reward can be given for reaching the destination efficiently. Negative rewards can be assigned for:

- Entering restricted airspace
- Collision
- Excessive energy consumption
- Taking unnecessarily long routes

Safety Penalties

A very large negative penalty should be assigned to entering restricted areas or colliding with obstacles. This makes unsafe actions much less attractive to the learned policy.

Decision

A constrained RL approach is suitable because the drone must optimize delivery efficiency while satisfying strict safety requirements.

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8. RL-Based Personalized Learning Paths

Understanding of Scenario

The university wants to personalize learning paths according to individual student performance and engagement.

State

The state can include:

- Current knowledge level
- Previous quiz scores
- Learning progress
- Time spent learning
- Completed topics
- Engagement level

Actions

The RL agent can recommend:

- A new topic
- Revision material
- Practice questions
- Advanced content
- Easier learning material

Reward Design

Positive rewards can be given for improved learning performance, completion of learning activities, and meaningful engagement. The reward should not depend only on time spent on the platform because this could encourage unnecessary usage.

Ethical Considerations

The system should:

- Treat students fairly.
- Avoid unfair recommendations.
- Protect student data.
- Provide transparent recommendations.
- Avoid creating excessive academic pressure.
- Allow human teachers to review important decisions.

Decision

RL can provide adaptive learning paths, but educational improvement and fairness should be prioritized over simply maximizing engagement.

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9. RL-Based Traffic Signal Control with Safety Constraints

Understanding of Scenario

The traffic management authority wants to use RL to reduce congestion at busy intersections while ensuring pedestrian and emergency vehicle safety.

State

The state can include:

- Number of vehicles in each lane
- Queue length
- Current traffic signal
- Pedestrian crossing status
- Emergency vehicle presence

Actions

The agent can:

- Change the traffic signal phase
- Extend green time
- Reduce green time
- Switch to another safe phase

Safety Constraints

Emergency vehicles should receive priority when detected. Pedestrian crossing phases should not be interrupted in an unsafe manner. Minimum green and red times should also be maintained.

Reward

Positive rewards can be given for:

- Reducing vehicle waiting time
- Increasing traffic flow
- Clearing queues

Negative rewards can be assigned for:

- Unsafe signal changes
- Pedestrian conflicts
- Delays to emergency vehicles

Decision

RL can reduce congestion by dynamically adjusting signal timing according to traffic conditions. However, safety constraints must have higher priority than congestion reduction.

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10. RL for Machine Scheduling Under Resource Limitations

Understanding of Scenario

The manufacturing plant needs to schedule machines efficiently while dealing with limited machines, workers, materials, and changing production requirements.

Traditional Optimization

Traditional optimization methods such as linear programming or scheduling algorithms generally optimize a predefined objective under known constraints. They can provide strong solutions when the production environment is relatively stable.

Constrained RL

Constrained RL learns a policy through interaction with the production environment. It can consider:

- Machine availability
- Resource limitations
- Production deadlines
- Machine failures
- Changing workloads

The agent receives rewards for efficient production and penalties for resource violations, delays, or machine conflicts.

Comparison

Traditional optimization is effective when the problem and constraints are well defined and relatively stable. Constrained RL is more adaptable when production conditions change continuously.

Decision

For a dynamic manufacturing environment, constrained RL can provide better adaptability because it can learn from changing conditions. However, traditional optimization may still be preferable for highly predictable scheduling problems where exact solutions are required.

RUBRICS FOR CALCULATION

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand